ASSIGNMENT -1
MODULE 1

COMPUTER NETWORKS
(PCCST501)

Submitted by,

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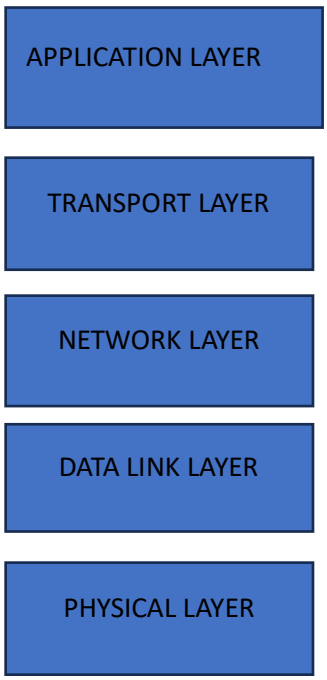
S5 CSE A

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How the Internet Delivers a Web Page
Website Chosen: YouTube (https://www.youtube.com)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. Each layer performs a specific function and provides services to the layer above it.



Application Layer

The application layer provides services directly to users and application programs. It enables communication between processes running on different hosts. Protocols such as HTTP, SMTP, FTP, DNS, and SSH operate at this layer.

Transport Layer

The transport layer provides process-to-process communication. It receives messages from the application layer, divides them into segments, and ensures reliable data delivery using TCP or fast communication using UDP.

Network Layer

The network layer is responsible for host-to-host communication. It uses the Internet Protocol (IP) to assign logical addresses and route packets through routers until they reach the destination.

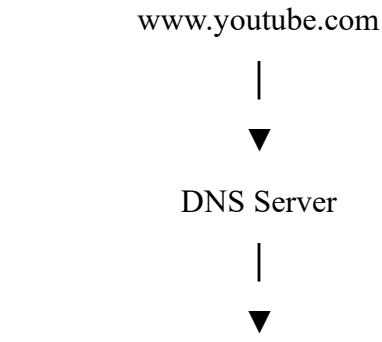
Data Link Layer The data-link layer transfers frames between adjacent devices on the same network. It adds MAC addresses and prepares data for transmission across the physical medium.

Physical Layer

The physical layer transmits bits as electrical, optical, or wireless signals over cables or air.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

When a user enters **www.youtube.com** into the browser, the browser cannot communicate using the domain name alone because communication on the Internet requires an IP address. The browser first contacts the **Domain Name System (DNS)**. DNS is an application-layer protocol that translates a human-readable domain name into its corresponding IP address. After receiving the IP address, the browser knows the location of YouTube's web server and can establish communication with it. DNS acts as a distributed database that maps names to IP addresses, making it easier for users to access Internet resources without remembering numerical addresses.



3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (Hyper Text Transfer Protocol) is the standard protocol used to access web pages. It follows the **client-server model**, where the browser acts as the client and the web server acts as the server.

The communication takes place as follows:

1. The browser sends an **HTTP request** to YouTube's web server.
2. The request usually uses the **GET** method to retrieve the required webpage.
3. The server processes the request.
4. The server sends an **HTTP response** containing the requested webpage.
5. The browser receives the response and displays the webpage.

HTTP request contains:

- Request line
- Header lines

- Blank line
- Optional body

An HTTP response contains:

- Status line
- Header lines
- Blank line
- Body

4. Explain How TCP Provides Reliable Communication for HTTP

TCP provides:

- **Reliable delivery**, by ensuring that all data reaches the destination correctly.
- **Error control**, which detects lost or damaged segments and retransmits them.
- **Flow control**, which prevents the sender from transmitting data faster than the receiver can process it.
- **Congestion control**, which reduces network congestion by controlling the transmission rate.

5. Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

YouTube follows the **Client–Server architecture**.

In this architecture, the **web browser** acts as the **client** and YouTube's **web server** acts as the **server**. The client sends an HTTP request to the server, and the server processes the request and returns the requested webpage as an HTTP response.

This architecture is suitable because the web server stores the web pages and provides services to multiple clients simultaneously.

6. Compare HTTP and FTP

HTTP	**FTP**
Used to access web pages on the World Wide Web.	Used to transfer files between computers.
Uses a single TCP connection for communication (persistent or nonpersistent).	Uses two TCP connections: one control connection and one data connection.
Mainly uses the GET and POST methods to request or send web resources.	Uses FTP commands such as RETR, STOR, LIST, and DELE for file operation

Primarily transfers HTML pages, images, CSS, and JavaScript files.	Transfers files of different formats between local and remote systems.

7.Explain How Cookies Improve User Experience with One Real-World Example

Cookies are used to remember information about a client. When a client visits a website, the server creates a cookie and sends it to the browser. The browser stores the cookie and includes it in future requests to the same server.

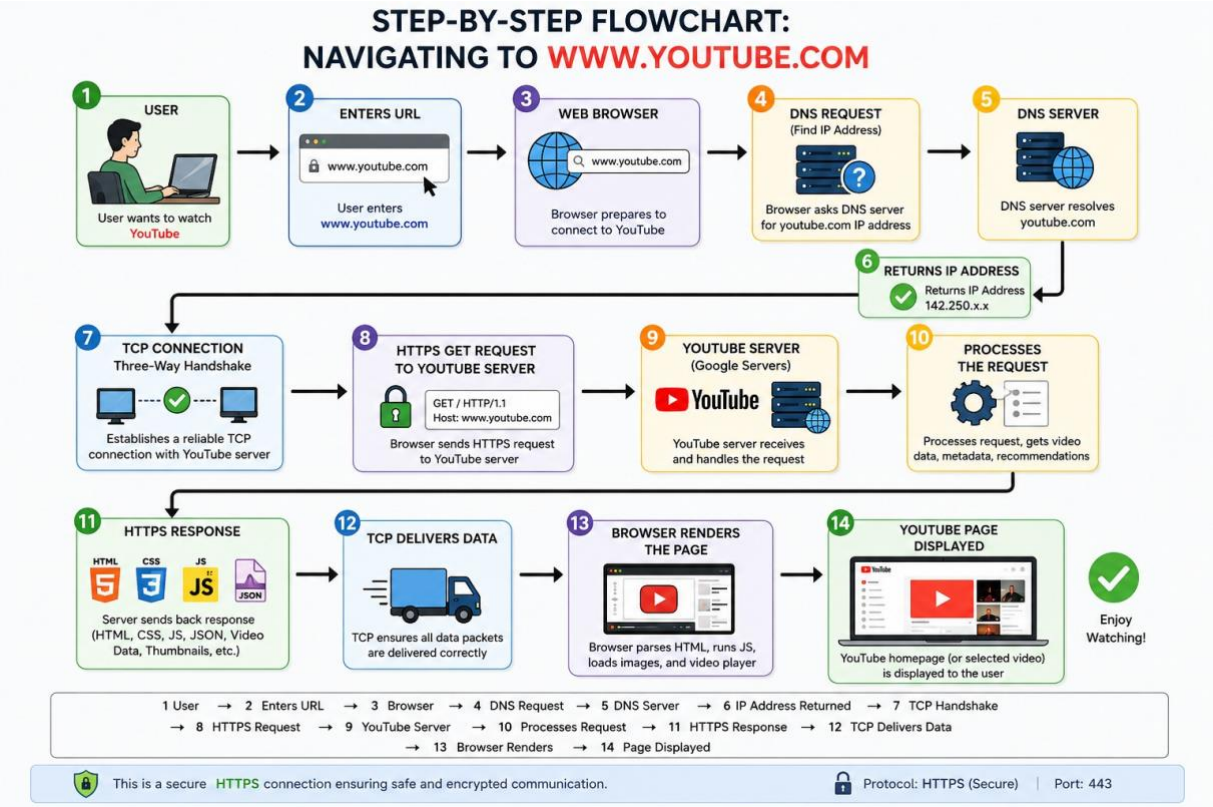
This allows the server to recognize the user without asking for the same information repeatedly.

Real-World Example:

When a user visits YouTube and signs in or adjusts their volume preferences, the server assigns a unique cookie ID to the browser. As the user navigates between different videos, the browser sends that cookie ID back to the server. YouTube uses this to keep the user logged in and preserve their volume levels across sessions, eliminating the need to re-login on every click.

Thus, cookies help websites remember user information and improve the browsing experience.

8. Flow Diagram Showing the Complete Communication



9.Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are essential for everyday Internet communication. They provide the services that users interact with directly, such as web browsing, file transfer, email, and remote access. Protocols like DNS convert domain names into IP addresses, making websites easy to access, while HTTP enables communication between web browsers and web servers to retrieve web pages. FTP allows efficient file transfer between systems, and cookies help improve the user experience by remembering user information and preferences. Working together with the other layers of the TCP/IP protocol suite, these protocols ensure reliable, efficient, and user-friendly communication. Understanding application layer protocols helps us understand how modern Internet services function seamlessly in our daily lives.